



Library Management system- Project Report

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About me: I am a diploma level student in IIT Madras BS in Data Science and application degree.

Description:

The LMS is a web app designed for easy e-book management. It uses Flask for backend and VueJS for frontend, offering smooth experiences for users and librarians. Inspired by Amazon and Book Depository, it efficiently handles section and e-book management, user authentication, and access control.

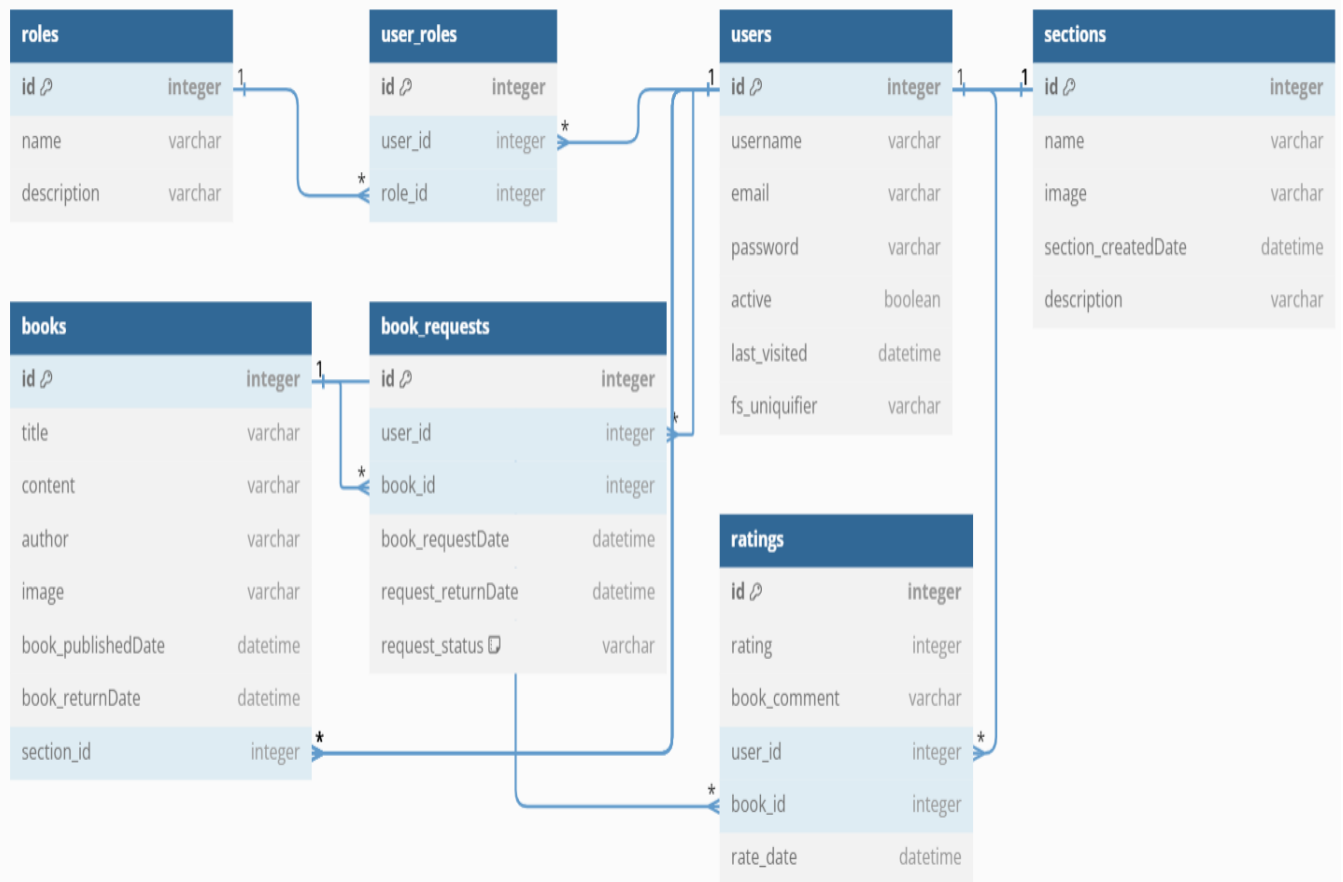
Key Features:

- I. **Authentication:** Secure login for users and librarians with username/email and password. Role-Based Access Control ensures proper permissions.
- II. **User Management:** Registration, login, and profile management for users. Librarians have exclusive admin access.
- III. **Section and E-book Management:** Librarians can add, edit, and remove sections and books. Each item has details like ID, name, and author.
- IV. **Issuing and Returning:** Users can request up to 5 e-books at a time, with access for a set period before automatic revocation.
- V. **Feedback and Ratings:** Users can leave feedback, enhancing the reading experience.
- VI. **Search:** Easy search functionality for sections and e-books based on various criteria.
- VII. **Scheduled Jobs:** Daily reminders for inactive users and monthly activity reports for user insights.

Technologies Used:

- I. Flask: Backend API development.
- II. VueJS: Frontend UI development.
- III. Jinja2 Templates: For dynamic content rendering (if required).
- IV. Bootstrap: For UI styling and responsiveness.
- V. SQLite: Database management.
- VI. Redis: Caching system for improved performance.
- VII. Celery: Task queue for managing background jobs.
- VIII. JWT (JSON Web Tokens): For secure authentication and authorization

Database schema:



API design:

The provided Flask RESTful API serves as the backbone of a library management system, offering endpoints to manage users, sections, and books. Utilizing JWT tokens for authentication, the API ensures secure access control to various functionalities. Users can perform CRUD operations on entities such as adding new users, sections, and books, as well as updating and deleting existing ones. Error handling is implemented to provide informative messages in JSON format for any encountered issues, enhancing usability. With scalability in mind, the API is designed to accommodate future enhancements and additions, making it a robust solution for efficient library management.

Architecture and features:

VueJs is used to make website faster and more responsive. The design is focused on creating a smooth experience for users by integrating Vue components seamlessly on a single page. Data retrieval speed is boosted by incorporating Redis and caching. Additionally, automated alert notifications and monthly reports using celery jobs is also integrated. These enhancements together create a sophisticated and strong system that prioritizes speed, efficiency, and automation to provide users with a great digital experience.

Video link:

To view short video of my project click here:

[PROJECT DEMONSTRATION VIDEO LINK](#)